

Skill Session 23

Task 1

To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance

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j_rakshitha@RAKSHITHA: ~  
j_rakshitha@RAKSHITHA:~$ nano session23_task1_pipeline.c  
j_rakshitha@RAKSHITHA:~$ gcc -Wall -Wextra session23_task1_pipeline.c -o session23_task1_pipeline  
j_rakshitha@RAKSHITHA:~$ ./session23_task1_pipeline  
Job 1 (ls) completed.  
Job 2 (grep) completed.  
-rw-r--r-- 1 j_rakshitha j_rakshitha 0 Sep 17 08:00 combined_output.txt  
-rw-r--r-- 1 j_rakshitha j_rakshitha 60 Sep 17 08:00 combined_error.txt  
-rw-r--r-- 1 j_rakshitha j_rakshitha 391 Aug 12 06:39 task1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 507 Sep 28 03:28 session22_task2_tests.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 653 Sep 17 08:29 fifo_client.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 746 Sep 17 09:14 proc_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 800 Sep 28 03:25 session22_task1_valgrind.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 828 Aug 18 05:45 week4_T2.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 933 Aug 12 06:41 task2.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 963 Sep 17 04:40 session10_pwd_exit.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 981 Aug 3 05:47 command_executor.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1218 Aug 12 06:44 task3.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1230 Sep 17 09:04 signal_handler.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1277 Sep 17 08:28 fifo_server.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1378 Aug 18 05:48 week4_T1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1432 Sep 17 09:20 session8_task1_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1518 Sep 17 09:10 memory_addresses.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1560 Sep 17 05:24 session12_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1733 Sep 28 03:38 session23_task1_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1786 Aug 12 06:52 task5.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1836 Sep 17 09:23 session8_task2_cow.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1924 Sep 17 08:08 session16_task1_background.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1926 Sep 17 06:49 session13_task2_output.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1970 Sep 17 04:42 session10_export.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2232 Sep 17 05:22 session11_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2306 Sep 27 15:20 session21_task1_modules.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2576 Sep 17 05:27 session12_multi_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2655 Sep 17 06:58 session14_task1_append.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2784 Sep 17 06:59 session14_task2_stderr.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2877 Sep 27 14:47 session19_task1_handlers.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2997 Sep 17 05:10 session11_history.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3000 Sep 27 15:13 session20_task2_process_groups.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3260 Sep 17 07:58 session15_task1_combined.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3533 Sep 17 08:03 session15_task2_complex.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3723 Sep 27 13:57 session17_task1_jobs.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3765 Sep 27 15:24 session21_task2_errors.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3814 Aug 12 06:50 task4.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 4116 Sep 17 08:10 session16_task2_jobtable.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 4406 Sep 27 15:05 session19_task2_sigint.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5051 Sep 27 14:38 session18_task2_background.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5548 Sep 27 14:20 session18_task1_signals.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5691 Sep 27 14:17 session17_task2_foreground.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 6419 Sep 27 15:08 session20_task1_sigtstp.c
```

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j_rakshitha@RAKSHITHA: ~  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16360 Aug  3 05:51 command_executor  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16560 Sep 17 09:24 session8_task2_cow  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16896 Sep 27 15:14 session20_task2_process_groups  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16936 Sep 27 14:39 session18_task2_background  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19304 Sep 17 09:14 proc_memory  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19616 Sep 17 08:29 fifo_client  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20224 Sep 17 08:08 session16_task1_background  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20880 Sep 17 07:58 session15_task1_combined  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 21032 Sep 17 08:03 session15_task2_complex  
drwxr-xr-x 5 j_rakshitha j_rakshitha 4096 Aug 18 05:21 ossp_project  
Job 3 (sort) completed.  
Large pipeline completed successfully.  
j_rakshitha@RAKSHITHA:~$ ./session23_task1_pipeline &  
[1] 998  
Job 1 (ls) completed.  
j_rakshitha@RAKSHITHA:~$ -rw-r--r-- 1 j_rakshitha j_rakshitha 0 Sep 17 08:00 combined_  
output.txt  
-rw-r--r-- 1 j_rakshitha j_rakshitha 60 Sep 17 08:00 combined_error.txt  
-rw-r--r-- 1 j_rakshitha j_rakshitha 391 Aug 12 06:39 task1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 507 Sep 28 03:28 session22_task2_tests.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 653 Sep 17 08:29 fifo_client.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 746 Sep 17 09:14 proc_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 800 Sep 28 03:25 session22_task1_valgrind.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 828 Aug 18 05:45 week4_T2.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 933 Aug 12 06:41 task2.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 963 Sep 17 04:40 session10_pwd_exit.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 981 Aug  3 05:47 command_executor.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1218 Aug 12 06:44 task3.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1230 Sep 17 09:04 signal_handler.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1277 Sep 17 08:28 fifo_server.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1378 Aug 18 05:48 week4_T1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1432 Sep 17 09:20 session8_task1_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1518 Sep 17 09:10 memory_addresses.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1560 Sep 17 05:24 session12_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1733 Sep 28 03:38 session23_task1_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1786 Aug 12 06:52 task5.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1836 Sep 17 09:23 session8_task2_cow.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1924 Sep 17 08:08 session16_task1_background.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1926 Sep 17 06:49 session13_task2_output.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1970 Sep 17 04:42 session10_export.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2232 Sep 17 05:22 session11_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2306 Sep 27 15:20 session21_task1_modules.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2576 Sep 17 05:27 session12_multi_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2655 Sep 17 06:58 session14_task1_append.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2784 Sep 17 06:59 session14_task2_stderr.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2877 Sep 27 14:47 session19_task1_handlers.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2997 Sep 17 05:10 session11_history.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3000 Sep 27 15:13 session20_task2_process_groups.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3260 Sep 17 07:58 session15_task1_combined.c
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j_rakshitha@RAKSHITHA: ~  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3533 Sep 17 08:03 session15_task2_complex.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3723 Sep 27 13:57 session17_task1_jobs.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3765 Sep 27 15:24 session21_task2_errors.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3814 Aug 12 06:50 task4.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 4116 Sep 17 08:10 session16_task2_jobtable.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 4406 Sep 27 15:05 session19_task2_sigint.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5051 Sep 27 14:38 session18_task2_background.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5548 Sep 27 14:20 session18_task1_signals.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5691 Sep 27 14:17 session17_task2_foreground.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 6419 Sep 27 15:08 session20_task1_sigstsp.c  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16360 Aug 3 05:51 command_executor  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16560 Sep 17 09:24 session8_task2_cow  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16896 Sep 27 15:14 session20_task2_process_groups  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16936 Sep 27 14:39 session18_task2_background  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19304 Sep 17 09:14 proc_memory  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19616 Sep 17 08:29 fifo_client  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20224 Sep 17 08:08 session16_task1_background  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20880 Sep 17 07:58 session15_task1_combined  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 21032 Sep 17 08:03 session15_task2_complex  
drwxr-xr-x 5 j_rakshitha j_rakshitha 4096 Aug 18 05:21 ossp_project  
Job 2 (grep) completed.  
Job 3 (sort) completed.  
Large pipeline completed successfully.  
./session23_task1_pipeline ./session23_task1_pipeline &  
[2] 1012  
[1] Done ./session23_task1_pipeline  
j_rakshitha@RAKSHITHA:~$ Job 1 (ls) completed.  
Job 2 (grep) completed.  
-rw-r--r-- 1 j_rakshitha j_rakshitha 0 Sep 17 08:00 combined_output.txt  
-rw-r--r-- 1 j_rakshitha j_rakshitha 60 Sep 17 08:00 combined_error.txt  
-rw-r--r-- 1 j_rakshitha j_rakshitha 391 Aug 12 06:39 task1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 507 Sep 28 03:28 session22_task2_tests.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 653 Sep 17 08:29 fifo_client.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 746 Sep 17 09:14 proc_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 800 Sep 28 03:25 session22_task1_valgrind.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 828 Aug 18 05:45 week4_T2.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 933 Aug 12 06:41 task2.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 963 Sep 17 04:40 session10_pwd_exit.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 981 Aug 3 05:47 command_executor.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1218 Aug 12 06:44 task3.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1230 Sep 17 09:04 signal_handler.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1277 Sep 17 08:28 fifo_server.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1378 Aug 18 05:48 week4_T1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1432 Sep 17 09:20 session8_task1_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1518 Sep 17 09:10 memory_addresses.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1560 Sep 17 05:24 session12_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1733 Sep 28 03:38 session23_task1_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1786 Aug 12 06:52 task5.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1836 Sep 17 09:23 session8_task2_cow.c
```

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j_rakshitha@RAKSHITHA: ~
-rw-r--r-- 1 j_rakshitha j_rakshitha 1924 Sep 17 08:08 session16_task1_background.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 1926 Sep 17 06:49 session13_task2_output.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 1970 Sep 17 04:42 session10_export.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2232 Sep 17 05:22 session11_pipeline.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2306 Sep 27 15:20 session21_task1_modules.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2576 Sep 17 05:27 session12_multi_pipe.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2655 Sep 17 06:58 session14_task1_append.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2784 Sep 17 06:59 session14_task2_stderr.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2877 Sep 27 14:47 session19_task1_handlers.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2997 Sep 17 05:10 session11_history.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 3000 Sep 27 15:13 session20_task2_process_groups.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 3260 Sep 17 07:58 session15_task1_combined.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 3533 Sep 17 08:03 session15_task2_complex.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 3723 Sep 27 13:57 session17_task1_jobs.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 3765 Sep 27 15:24 session21_task2_errors.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 3814 Aug 12 06:50 task4.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 4116 Sep 17 08:10 session16_task2_jobtable.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 4406 Sep 27 15:05 session19_task2_sigint.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 5051 Sep 27 14:38 session18_task2_background.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 5548 Sep 27 14:20 session18_task1_signals.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 5691 Sep 27 14:17 session17_task2_foreground.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 6419 Sep 27 15:08 session20_task1_sigtstp.c
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16360 Aug 3 05:51 command_executor
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16560 Sep 17 09:24 session8_task2_cow
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16896 Sep 27 15:14 session20_task2_process_groups
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16936 Sep 27 14:39 session18_task2_background
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19304 Sep 17 09:14 proc_memory
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19616 Sep 17 08:29 fifo_client
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20224 Sep 17 08:08 session16_task1_background
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20880 Sep 17 07:58 session15_task1_combined
-rwxr-xr-x 1 j_rakshitha j_rakshitha 21032 Sep 17 08:03 session15_task2_complex
drwxr-xr-x 5 j_rakshitha j_rakshitha 4096 Aug 18 05:21 ossp_project
Job 3 (sort) completed.
Large pipeline completed successfully.
./session23_task1_pipeline ./session23_task1_pipeline &
[3] 1026
[2] Done ./session23_task1_pipeline
j_rakshitha@RAKSHITHA:~$ Job 1 (ls) completed.
Job 2 (grep) completed.
-rw-r--r-- 1 j_rakshitha j_rakshitha 0 Sep 17 08:00 combined_output.txt
-rw-r--r-- 1 j_rakshitha j_rakshitha 60 Sep 17 08:00 combined_error.txt
-rw-r--r-- 1 j_rakshitha j_rakshitha 391 Aug 12 06:39 task1.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 507 Sep 28 03:28 session22_task2_tests.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 653 Sep 17 08:29 fifo_client.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 746 Sep 17 09:14 proc_memory.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 800 Sep 28 03:25 session22_task1_valgrind.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 828 Aug 18 05:45 week4_T2.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 933 Aug 12 06:41 task2.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 963 Sep 17 04:40 session10_pwd_exit.c

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j_rakshitha@RAKSHITHA: ~  
-rw-r--r-- 1 j_rakshitha j_rakshitha 981 Aug 3 05:47 command_executor.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1218 Aug 12 06:44 task3.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1230 Sep 17 09:04 signal_handler.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1277 Sep 17 08:28 fifo_server.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1378 Aug 18 05:48 week4_T1.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1432 Sep 17 09:20 session8_task1_memory.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1518 Sep 17 09:10 memory_addresses.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1560 Sep 17 05:24 session12_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1733 Sep 28 03:38 session23_task1_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1786 Aug 12 06:52 task5.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1836 Sep 17 09:23 session8_task2_cow.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1924 Sep 17 08:08 session16_task1_background.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1926 Sep 17 06:49 session13_task2_output.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 1970 Sep 17 04:42 session10_export.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2232 Sep 17 05:22 session11_pipeline.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2306 Sep 27 15:20 session21_task1_modules.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2576 Sep 17 05:27 session12_multi_pipe.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2655 Sep 17 06:58 session14_task1_append.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2784 Sep 17 06:59 session14_task2_stderr.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2877 Sep 27 14:47 session19_task1_handlers.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 2997 Sep 17 05:10 session11_history.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3000 Sep 27 15:13 session20_task2_process_groups.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3260 Sep 17 07:58 session15_task1_combined.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3533 Sep 17 08:03 session15_task2_complex.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3723 Sep 27 13:57 session17_task1_jobs.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3765 Sep 27 15:24 session21_task2_errors.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 3814 Aug 12 06:50 task4.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 4116 Sep 17 08:10 session16_task2_jobtable.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 4406 Sep 27 15:05 session19_task2_sigint.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5051 Sep 27 14:38 session18_task2_background.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5548 Sep 27 14:20 session18_task1_signals.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 5691 Sep 27 14:17 session17_task2_foreground.c  
-rw-r--r-- 1 j_rakshitha j_rakshitha 6419 Sep 27 15:08 session20_task1_sigstsp.c  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16360 Aug 3 05:51 command_executor  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16560 Sep 17 09:24 session8_task2_cow  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16896 Sep 27 15:14 session20_task2_process_groups  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 16936 Sep 27 14:39 session18_task2_background  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19304 Sep 17 09:14 proc_memory  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 19616 Sep 17 08:29 fifo_client  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20224 Sep 17 08:08 session16_task1_background  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 20880 Sep 17 07:58 session15_task1_combined  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 21032 Sep 17 08:03 session15_task2_complex  
drwxr-xr-x 5 j_rakshitha j_rakshitha 4096 Aug 18 05:21 ossp_project  
Job 3 (sort) completed.  
Large pipeline completed successfully.  
ps -ef | grep session23_tps -ef | grep session23_task1_pipeline | grep -v grep  
[3]+ Done ./session23_task1_pipeline  
j_rakshitha@RAKSHITHA:~$ echo $?  
1
```

```
j_rakshitha@RAKSHITHA: ~ × + ▾
GNU nano 8.7.1 session23_task1_pipeline.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    int pipe1[2];
    int pipe2[2];

    if (pipe(pipe1) == -1 || pipe(pipe2) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t p1 = fork();

    if (p1 == -1)
    {
        perror("fork");
        return 1;
    }

    if (p1 == 0)
    {
        dup2(pipe1[1], STDOUT_FILENO);

        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[0]);
        close(pipe2[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
        exit(1);
    }

    pid_t p2 = fork();

    if (p2 == -1)
    {
        perror("fork");
        return 1;
    }

    [ Read 106 lines ]
    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~ × + v
GNU nano 8.7.1 session23_task1_pipeline.c
}

if (p2 == 0)
{
    dup2(pipe1[0], STDIN_FILENO);
    dup2(pipe2[1], STDOUT_FILENO);

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp grep");
    exit(1);
}

pid_t p3 = fork();

if (p3 == -1)
{
    perror("fork");
    return 1;
}

if (p3 == 0)
{
    dup2(pipe2[0], STDIN_FILENO);

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    execlp("sort", "sort", NULL);

    perror("execlp sort");
    exit(1);
}

close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
int status;

waitpid(p1, &status, 0);
printf("Job 1 (ls) completed.\n");

waitpid(p2, &status, 0);
printf("Job 2 (grep) completed.\n");

waitpid(p3, &status, 0);
printf("Job 3 (sort) completed.\n");

printf("Large pipeline completed successfully.\n");

return 0;
}
```

Task 2

To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation

```
j_rakshitha@RAKSHITHA: ~  
j_rakshitha@RAKSHITHA:~$ nano session23_task2_architecture.txt  
j_rakshitha@RAKSHITHA:~$ cat session23_task2_architecture.txt  
PRACTICAL SKILL SESSION 23 - TASK 2  
=====
```

SYSTEM ARCHITECTURE

The program implements a three-stage process pipeline.

Execution Flow:

```
graph TD
    Parent[Parent Process] --> Child1[Child 1: ls -l]
    Child1 -- pipe1 --> Child2[Child 2: grep .c]
    Child2 -- pipe2 --> Child3[Child 3: sort]
    Child3 --> Wait[Wait for all child processes]
    Wait --> Done[Pipeline completed]
```

Parent Process

- +----> Child 1: ls -l
 - ↓
pipe1
- +----> Child 2: grep .c
 - ↓
pipe2
- +----> Child 3: sort

↓
Wait for all child processes

↓
Pipeline completed

EXECUTION PATH

1. The parent creates two pipes.
2. The parent creates the first child process.
3. Child 1 redirects stdout to pipe1.
4. Child 1 executes "ls -l".
5. The parent creates the second child.
6. Child 2 reads from pipe1.
7. Child 2 redirects stdout to pipe2.
8. Child 2 executes "grep .c".
9. The parent creates the third child.
10. Child 3 reads from pipe2.
11. Child 3 executes "sort".
12. The parent closes unused pipe descriptors.
13. The parent waits for all three children.

```
j_rakshitha@RAKSHITHA: ~ × + v
14. The pipeline reports successful completion.

IMPORTANT SYSTEM CALLS
-----

pipe()
    Creates communication channels between processes.

fork()
    Creates child processes.

dup2()
    Redirects standard input/output to pipe descriptors.

execlp()
    Replaces a child process with the required command.

close()
    Closes unused file descriptors.

waitpid()
    Allows the parent to wait for child processes to finish.

DESIGN DECISIONS
-----

1. Two pipes are used because three commands require two communication channels.
2. Each command runs in a separate child process.
3. dup2() is used for input/output redirection.
4. Unused pipe descriptors are closed to avoid descriptor leaks and incorrect pipeline behavior.
5. The parent waits for every child process before reporting completion.
6. Error checking is performed for pipe() and fork() operations.

VALIDATION
-----

The pipeline was compiled using:

gcc -Wall -Wextra session23_task1_pipeline.c -o session23_task1_pipeline
```

 j_rakshitha@RAKSHITHA: ~

The program was executed successfully.

The following stages completed:

Job 1 (ls) completed.

```
Job 2 (grep) completed.
```

Job 3 (sort) completed.

Multiple background executions were also tested.

```
No session23_task1_pipeline process remained after completion.
```

CONCLUSION

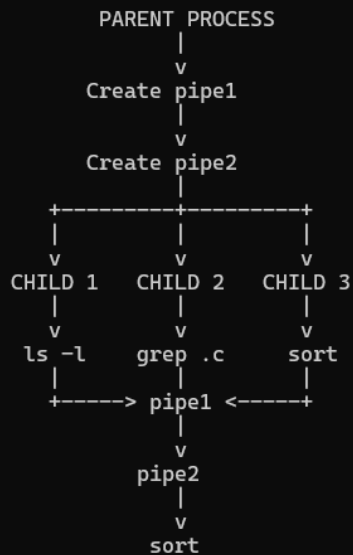
The architecture demonstrates process creation, inter-process communication, pipe-based execution, command execution, process synchronization, and cleanup.

```
j_rakshitha@RAKSHITHA:~$ nano session23_task2_flow.txt
```

```
j_rakshitha@RAKSHITHA:~$ cat session23_task2_flow.txt
```

SESSION 23 – TASK 2

PIPELINE EXECUTION FLOW



```
j_rakshitha@RAKSHITHA: ~ × + ▾

      |
      v
Wait for all children
      |
      v
Pipeline completed

SYSTEM CALL FLOW
=====

pipe()
|
v
fork()
|
+--> Child 1 --> dup2() --> execlp("ls")
|
+--> Child 2 --> dup2() --> execlp("grep")
|
+--> Child 3 --> dup2() --> execlp("sort")
|
v
close()
|
v
waitpid()
|
v
Program completion
j_rakshitha@RAKSHITHA:~$ ls -l session23_task1_pipeline.c session23_task2_architecture.txt
session23_task2_flow.txt
-rw-r--r-- 1 j_rakshitha j_rakshitha 1733 Sep 28 03:38 session23_task1_pipeline.c
-rw-r--r-- 1 j_rakshitha j_rakshitha 2499 Sep 28 03:44 session23_task2_architecture.txt
-rw-r--r-- 1 j_rakshitha j_rakshitha 1159 Sep 28 03:47 session23_task2_flow.txt
j_rakshitha@RAKSHITHA:~$ cat session23_task2_architecture.txt
cat session23_task2_flow.txt
PRACTICAL SKILL SESSION 23 - TASK 2
=====

SYSTEM ARCHITECTURE
-----

The program implements a three-stage process pipeline.

Execution Flow:

Parent Process
|
```

```
j_rakshitha@RAKSHITHA: ~  
|  
+----> Child 1: ls -l  
|  
| v  
| pipe1  
|  
+----> Child 2: grep .c  
|  
| v  
| pipe2  
|  
+----> Child 3: sort  
|  
| v  
Wait for all child processes  
|  
| v  
Pipeline completed  
  
EXECUTION PATH  
-----  
  
1. The parent creates two pipes.  
2. The parent creates the first child process.  
3. Child 1 redirects stdout to pipe1.  
4. Child 1 executes "ls -l".  
5. The parent creates the second child.  
6. Child 2 reads from pipe1.  
7. Child 2 redirects stdout to pipe2.  
8. Child 2 executes "grep .c".  
9. The parent creates the third child.  
10. Child 3 reads from pipe2.  
11. Child 3 executes "sort".  
12. The parent closes unused pipe descriptors.  
13. The parent waits for all three children.  
14. The pipeline reports successful completion.  
  
IMPORTANT SYSTEM CALLS  
-----  
  
pipe()  
    Creates communication channels between processes.  
  
fork()  
    Creates child processes.  
  
dup2()
```

Redirects standard input/output to pipe descriptors.

`execlp()`
Replaces a child process with the required command.

`close()`
Closes unused file descriptors.

`waitpid()`
Allows the parent to wait for child processes to finish.

DESIGN DECISIONS

1. Two pipes are used because three commands require two communication channels.
2. Each command runs in a separate child process.
3. `dup2()` is used for input/output redirection.
4. Unused pipe descriptors are closed to avoid descriptor leaks and incorrect pipeline behavior.
5. The parent waits for every child process before reporting completion.
6. Error checking is performed for `pipe()` and `fork()` operations.

VALIDATION

The pipeline was compiled using:

```
gcc -Wall -Wextra session23_task1_pipeline.c -o session23_task1_pipeline
```

The program was executed successfully.

The following stages completed:

Job 1 (ls) completed.
Job 2 (grep) completed.
Job 3 (sort) completed.

Multiple background executions were also tested.

No `session23_task1_pipeline` process remained after completion.

```
j_rakshitha@RAKSHITHA: ~ × + v
CONCLUSION
=====

The architecture demonstrates process creation, inter-process
communication, pipe-based execution, command execution, process
synchronization, and cleanup.
=====
SESSION 23 - TASK 2
PIPELINE EXECUTION FLOW
=====

      PARENT PROCESS
      |
      v
    Create pipe1
      |
      v
    Create pipe2
      |
+-----+-----+
|         |         |
|         v         v         v
|      CHILD 1    CHILD 2    CHILD 3
|         |         |         |
|         v         v         v
|      ls -l      grep .c      sort
|         |         |         |
+-----> pipe1 <-----+
|
|         v
|      pipe2
|         |
|         v
|      sort
|         |
|         v
|    Wait for all children
|         |
|         v
|    Pipeline completed

SYSTEM CALL FLOW
=====

pipe()
|
v
```

```
fork()
|
+--> Child 1 --> dup2() --> execlp("ls")
|
+--> Child 2 --> dup2() --> execlp("grep")
|
+--> Child 3 --> dup2() --> execlp("sort")
|
v
close()
|
v
waitpid()
|
v
Program completion
j_rakshitha@RAKSHITHA:~$ |
```

```
j_rakshitha@RAKSHITHA: ~ × + ▾
GNU nano 8.7.1 session23_task2_architecture.txt
PRACTICAL SKILL SESSION 23 - TASK 2
=====

SYSTEM ARCHITECTURE
-----

The program implements a three-stage process pipeline.

Execution Flow:

Parent Process
|
+----> Child 1: ls -l
|      |
|      v
|      pipe1
|      |
+----> Child 2: grep .c
|      |
|      v
|      pipe2
|      |
+----> Child 3: sort
|
v
Wait for all child processes
|
v
Pipeline completed

EXECUTION PATH
-----

1. The parent creates two pipes.
2. The parent creates the first child process.
3. Child 1 redirects stdout to pipe1.
4. Child 1 executes "ls -l".
5. The parent creates the second child.
6. Child 2 reads from pipe1.
7. Child 2 redirects stdout to pipe2.
8. Child 2 executes "grep .c".
9. The parent creates the third child.
10. Child 3 reads from pipe2.
11. Child 3 executes "sort".
12. The parent closes unused pipe descriptors.
13. The parent waits for all three children.
14. The pipeline reports successful completion.

IMPORTANT SYSTEM CALLS

[ Read 116 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
j_rakshitha@RAKSHITHA: ~ × + v
GNU nano 8.7.1 session23_task2_architecture.txt

pipe()
    Creates communication channels between processes.

fork()
    Creates child processes.

dup2()
    Redirects standard input/output to pipe descriptors.

execlp()
    Replaces a child process with the required command.

close()
    Closes unused file descriptors.

waitpid()
    Allows the parent to wait for child processes to finish.

DESIGN DECISIONS
-----

1. Two pipes are used because three commands require two communication
   channels.

2. Each command runs in a separate child process.

3. dup2() is used for input/output redirection.

4. Unused pipe descriptors are closed to avoid descriptor leaks and
   incorrect pipeline behavior.

5. The parent waits for every child process before reporting completion.

6. Error checking is performed for pipe() and fork() operations.

VALIDATION
-----

The pipeline was compiled using:

gcc -Wall -Wextra session23_task1_pipeline.c -o session23_task1_pipeline

The program was executed successfully.

The following stages completed:

Job 1 (ls) completed.
Job 2 (grep) completed.

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
Job 3 (sort) completed.

Multiple background executions were also tested.

No session23_task1_pipeline process remained after completion.
```

CONCLUSION

The architecture demonstrates process creation, inter-process communication, pipe-based execution, command execution, process synchronization, and cleanup.

```
j_rakshitha@RAKSHITHA: ~  
GNU nano 8.7.1 session23_task2_flow.txt  
=====br/>SESSION 23 - TASK 2  
PIPELINE EXECUTION FLOW  
=====br/>  
PARENT PROCESS  
|  
v  
Create pipe1  
|  
v  
Create pipe2  
|  
+-----+-----+  
|         |         |  
v         v         v  
CHILD 1   CHILD 2   CHILD 3  
|         |         |  
v         v         v  
ls -l     grep .c    sort  
|         |         |  
+-----> pipe1 <-----+  
|  
v  
pipe2  
|  
v  
sort  
|  
v  
Wait for all children  
|  
v  
Pipeline completed  
  
SYSTEM CALL FLOW  
=====br/>  
pipe()  
|  
v  
fork()  
|  
+--> Child 1 --> dup2() --> execlp("ls")  
|  
+--> Child 2 --> dup2() --> execlp("grep")  
|  
+--> Child 3 --> dup2() --> execlp("sort")  
|  
v  
[ Read 58 lines ]  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify  
^C Location   M-U Undo  
^_/ Go To Line M-E Redo
```

```
|  
v  
close()  
|  
v  
waitpid()  
|  
v  
Program completion  
|
```

Skill Session 24

Task 1

To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
j_rakshitha@RAKSHITHA:~$ mkdir -p ossp_project/src ossp_project/docs ossp_project/build
j_rakshitha@RAKSHITHA:~$ cp session23_task1_pipeline.c ossp_project/src/
cp session23_task2_architecture.txt ossp_project/docs/
cp session23_task2_flow.txt ossp_project/docs/
j_rakshitha@RAKSHITHA:~$ find ossp_project -type f
ossp_project/Makefile
ossp_project/build/tokenizer
ossp_project/build/wait_waitpid
ossp_project/build/child_exec
ossp_project/build/dquote_demo
ossp_project/build/history_shell
ossp_project/build/memory_demo
ossp_project/build/session9_task1_cd
ossp_project/build/file_copy
ossp_project/build/escape_sequence_demo
ossp_project/build/syntax_parser
ossp_project/build/pipe_ls_grep
ossp_project/build/escape_demo
ossp_project/build/double_quote_demo
ossp_project/build/main
ossp_project/build/process_states
ossp_project/build/parser
ossp_project/build/process_demo
ossp_project/build/myshell
ossp_project/build/pipe_producer_consumer
ossp_project/build/squote_demo
ossp_project/build/session9_task2_dispatch
ossp_project/build/session10_task1
ossp_project/build/pipeline_demo
ossp_project/build/file_filter_pipe
ossp_project/build/path_resolver
ossp_project/build/builtin_shell
ossp_project/build/keyboard_shell
ossp_project/build/fork_exec
ossp_project/build/single_quote_demo
ossp_project/build/zombie_demo
ossp_project/build/waitpid_monitor
ossp_project/build/variable_expand
ossp_project/build/child_process_demo
ossp_project/src/parser.c
ossp_project/src/session9_task2_dispatch.c
ossp_project/src/history_shell.c
ossp_project/src/file_copy.c
ossp_project/src/pipe_ls_grep.c
ossp_project/src/single_quotes.c+
ossp_project/src/session10_task1.c
ossp_project/src/escape_demo.c
ossp_project/src/double_quote_demo.c
ossp_project/src/squote_demo.c
```

```
j_rakshitha@RAKSHITHA: ~/o × + v
ossp_project/src/session23_task1_pipeline.c
ossp_project/src/tokenizer.c
ossp_project/src/escape_sequence_demo.c
ossp_project/src/waitpid_monitor.c
ossp_project/src/process_states.c
ossp_project/src/file_filter_pipe.c
ossp_project/src/builtin_shell.c
ossp_project/src/pipeline_demo.c
ossp_project/src/session10_task1.c.save
ossp_project/src/process_demo.c
ossp_project/src/memory_demo.c
ossp_project/src/myshell.c
ossp_project/src/main.c
ossp_project/src/variable_expand.c
ossp_project/src/zombie_demo.c
ossp_project/src/keyboard_shell.c
ossp_project/src/session9_task1_cd.c
ossp_project/src/fork_exec.c
ossp_project/src/single_quote_demo.c
ossp_project/src/dquote_demo.c
ossp_project/src/path_resolver.c
ossp_project/src/wait_waitpid.c
ossp_project/src/child_exec.c
ossp_project/src/child_process_demo.c
ossp_project/src/syntax_parser.c
ossp_project/src/pipe_producer_consumer.c
ossp_project/copied.txt
ossp_project/sample.txt
ossp_project/docs/session23_task2_architecture.txt
ossp_project/docs/session23_task2_flow.txt
j_rakshitha@RAKSHITHA:~$ cat ossp_project/Makefile
CC = gcc
CFLAGS = -Wall -Wextra -g

TARGET = build/main
SRC = src/main.c

all:
    $(CC) $(CFLAGS) $(SRC) -o $(TARGET)

clean:
    rm -f $(TARGET)

run: all
    ./${TARGET}
j_rakshitha@RAKSHITHA:~$ cd ossp_project
make
gcc -Wall -Wextra -g src/main.c -o build/main
j_rakshitha@RAKSHITHA:~/ossp_project$ ls -l build/main
```

```
j_rakshitha@RAKSHITHA: ~/o  × + ∨  
-rwxr-xr-x 1 j_rakshitha j_rakshitha 17064 Sep 28 03:54 build/main  
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/main  
OSSP Project Started Successfully!  
j_rakshitha@RAKSHITHA:~/ossp_project$ nano README.md  
j_rakshitha@RAKSHITHA:~/ossp_project$ ^C  
j_rakshitha@RAKSHITHA:~/ossp_project$ cat README.md  
OSSP PROJECT  
=====
```

Operating Systems and System Programming Practical Project

PROJECT STRUCTURE

src/
Source code files

build/
Compiled executable files

docs/
Project documentation and architecture notes

Makefile
Build and run instructions

BUILD

To compile the main project:

make

RUN

To run the project:

./build/main

CLEAN

To remove the main compiled executable:

make clean

```
j_rakshitha@RAKSHITHA: ~/o × + v
CURRENT VALIDATION
-----

The project was successfully compiled using:

make

The executable was generated at:

build/main

The program was executed successfully and displayed:

OSSP Project Started Successfully!

DOCUMENTATION
-----

Session 23 Task 2 documentation is stored in:

docs/session23_task2_architecture.txt
docs/session23_task2_flow.txt

SOURCE
-----

The project contains OSSP practical source files organized
under the src directory.
j_rakshitha@RAKSHITHA:~/ossp_project$ find . -maxdepth 2 -type f | sort
./Makefile
./README.md
./build/builtin_shell
./build/child_exec
./build/child_process_demo
./build/double_quote_demo
./build/dquote_demo
./build/escape_demo
./build/escape_sequence_demo
./build/file_copy
./build/file_filter_pipe
./build/fork_exec
./build/history_shell
./build/keyboard_shell
./build/main
./build/memory_demo
./build/myshell
```

j_rakshitha@RAKSHITHA: ~/o

```
./build/parser
./build/path_resolver
./build/pipe_ls_grep
./build/pipe_producer_consumer
./build/pipeline_demo
./build/process_demo
./build/process_states
./build/session10_task1
./build/session9_task1_cd
./build/session9_task2_dispatch
./build/single_quote_demo
./build/squote_demo
./build/syntax_parser
./build/tokenizer
./build/variable_expand
./build/wait_waitpid
./build/waitpid_monitor
./build/zombie_demo
./copied.txt
./docs/session23_task2_architecture.txt
./docs/session23_task2_flow.txt
./sample.txt
./src/builtin_shell.c
./src/child_exec.c
./src/child_process_demo.c
./src/double_quote_demo.c
./src/dquote_demo.c
./src/escape_demo.c
./src/escape_sequence_demo.c
./src/file_copy.c
./src/file_filter_pipe.c
./src/fork_exec.c
./src/history_shell.c
./src/keyboard_shell.c
./src/main.c
./src/memory_demo.c
./src/myshell.c
./src/parser.c
./src/path_resolver.c
./src/pipe_ls_grep.c
./src/pipe_producer_consumer.c
./src/pipeline_demo.c
./src/process_demo.c
./src/process_states.c
./src/session10_task1.c
./src/session10_task1.c.save
./src/session23_task1_pipeline.c
./src/session9_task1_cd.c
./src/session9_task2_dispatch.c
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
./src/single_quote_demo.c
./src/single_quotes.c+
./src/squote_demo.c
./src/syntax_parser.c
./src/tokenizer.c
./src/variable_expand.c
./src/wait_waitpid.c
./src/waitpid_monitor.c
./src/zombie_demo.c
j_rakshitha@RAKSHITHA:~/ossp_project$ git status
fatal: not a git repository (or any of the parent directories): .git
j_rakshitha@RAKSHITHA:~/ossp_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/j_rakshitha/ossp_project/.git/
j_rakshitha@RAKSHITHA:~/ossp_project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Makefile
    README.md
    build/
    copied.txt
    docs/
    sample.txt
    src/

nothing added to commit but untracked files present (use "git add" to track)
j_rakshitha@RAKSHITHA:~/ossp_project$ git add .
j_rakshitha@RAKSHITHA:~/ossp_project$ git commit -m "Initial OSSP project organization"
Author identity unknown

*** Please tell me who you are.

Run
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾

git config --global user.email "you@example.com"
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <j_rakshitha@RAKSHITHA.localdomain>) not allowed
j_rakshitha@RAKSHITHA:~/ossp_project$ git tag v1.0
fatal: Failed to resolve 'HEAD' as a valid ref.
j_rakshitha@RAKSHITHA:~/ossp_project$ git tag
j_rakshitha@RAKSHITHA:~/ossp_project$ git config --global user.name "Rakshitha"
j_rakshitha@RAKSHITHA:~/ossp_project$ git config --global user.email "rakshitha200804@gmail.com"
j_rakshitha@RAKSHITHA:~/ossp_project$ git config --global user.name
git config --global user.email
Rakshitha
rakshitha200804@gmail.com
j_rakshitha@RAKSHITHA:~/ossp_project$ git commit -m "Initial OSSP project organization"
[master (root-commit) a88ad7f] Initial OSSP project organization
75 files changed, 3495 insertions(+)
create mode 100644 Makefile
create mode 100644 README.md
create mode 100755 build/builtin_shell
create mode 100755 build/child_exec
create mode 100755 build/child_process_demo
create mode 100755 build/double_quote_demo
create mode 100755 build/dquote_demo
create mode 100755 build/escape_demo
create mode 100755 build/escape_sequence_demo
create mode 100755 build/file_copy
create mode 100755 build/file_filter_pipe
create mode 100755 build/fork_exec
create mode 100755 build/history_shell
create mode 100755 build/keyboard_shell
create mode 100755 build/main
create mode 100755 build/memory_demo
create mode 100755 build/myshell
create mode 100755 build/parser
create mode 100755 build/path_resolver
create mode 100755 build/pipe_ls_grep
create mode 100755 build/pipe_producer_consumer
create mode 100755 build/pipeline_demo
create mode 100755 build/process_demo
create mode 100755 build/process_states
create mode 100755 build/session10_task1
create mode 100755 build/session9_task1_cd
create mode 100755 build/session9_task2_dispatch
create mode 100755 build/single_quote_demo
create mode 100755 build/squote_demo
```

```
j_rakshitha@RAKSHITHA: ~/o x + v
create mode 100755 build/syntax_parser
create mode 100755 build/tokenizer
create mode 100755 build/variable_expand
create mode 100755 build/wait_waitpid
create mode 100755 build/waitpid_monitor
create mode 100755 build/zombie_demo
create mode 100644 copied.txt
create mode 100644 docs/session23_task2_architecture.txt
create mode 100644 docs/session23_task2_flow.txt
create mode 100644 sample.txt
create mode 100644 src/builtin_shell.c
create mode 100644 src/child_exec.c
create mode 100644 src/child_process_demo.c
create mode 100644 src/double_quote_demo.c
create mode 100644 src/dquote_demo.c
create mode 100644 src/escape_demo.c
create mode 100644 src/escape_sequence_demo.c
create mode 100644 src/file_copy.c
create mode 100644 src/file_filter_pipe.c
create mode 100644 src/fork_exec.c
create mode 100644 src/history_shell.c
create mode 100644 src/keyboard_shell.c
create mode 100644 src/main.c
create mode 100644 src/memory_demo.c
create mode 100644 src/myshell.c
create mode 100644 src/parser.c
create mode 100644 src/path_resolver.c
create mode 100644 src/pipe_ls_grep.c
create mode 100644 src/pipe_producer_consumer.c
create mode 100644 src/pipeline_demo.c
create mode 100644 src/process_demo.c
create mode 100644 src/process_states.c
create mode 100644 src/session10_task1.c
create mode 100644 src/session10_task1.c.save
create mode 100644 src/session23_task1_pipeline.c
create mode 100644 src/session9_task1_cd.c
create mode 100644 src/session9_task2_dispatch.c
create mode 100644 src/single_quote_demo.c
create mode 100644 src/single_quotes.c+
create mode 100644 src/squote_demo.c
create mode 100644 src/syntax_parser.c
create mode 100644 src/tokenizer.c
create mode 100644 src/variable_expand.c
create mode 100644 src/wait_waitpid.c
create mode 100644 src/waitpid_monitor.c
create mode 100644 src/zombie_demo.c
j_rakshitha@RAKSHITHA:~/ossp_project$ git tag v1.0
j_rakshitha@RAKSHITHA:~/ossp_project$ git status
git tag
```

```
j_rakshitha@RAKSHITHA:~/ossp_project$ git status
git tag
On branch master
nothing to commit, working tree clean
v1.0
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
j_rakshitha@RAKSHITHA: ~/o × + v
GNU nano 8.7.1 README.md
OSSP PROJECT
=====

Operating Systems and System Programming Practical Project

PROJECT STRUCTURE
-----

src/
  Source code files
build/
  Compiled executable files
docs/
  Project documentation and architecture notes
Makefile
  Build and run instructions

BUILD
-----

To compile the main project:

make

RUN
---

To run the project:

./build/main

CLEAN
-----

To remove the main compiled executable:

make clean

CURRENT VALIDATION
-----

The project was successfully compiled using:

make
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo
```

The executable was generated at:

build/main

The program was executed successfully and displayed:

OSSP Project Started Successfully!

DOCUMENTATION

Session 23 Task 2 documentation is stored in:

docs/session23_task2_architecture.txt
docs/session23_task2_flow.txt

SOURCE

The project contains OSSP practical source files organized under the src directory.

|

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

^C Location
^/ Go To Line

M-U Undo
M-E Redo

Task 2

To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review

```
j_rakshitha@RAKSHITHA: ~/o × + v
j_rakshitha@RAKSHITHA:~/ossp_project$ nano session24_task2_demo.txt
j_rakshitha@RAKSHITHA:~/ossp_project$ cat session24_task2_demo.txt
PRACTICAL SKILL SESSION 24 - TASK 2
=====

FINAL PROJECT DEMONSTRATION
-----

1. BUILD VERIFICATION

Command:
make

Result:
Project compiled successfully.

2. MAIN PROGRAM

Command:
./build/main

Result:
OSSP Project Started Successfully!

3. PIPELINE DEMONSTRATION

Program:
session23_task1_pipeline

Pipeline:
ls -l | grep .c | sort

Result:
Three-stage pipeline executed successfully.

4. SIGNAL DEMONSTRATION

The OSSP practical work included signal handling for:

SIGINT
SIGTERM
SIGUSR1
SIGTSTP
SIGCONT

Signals were tested with foreground and background processes.

5. PROCESS MANAGEMENT

The project work demonstrated:
```

```
j_rakshitha@RAKSHITHA: ~/o × + v
- Process groups
- Foreground execution
- Background execution
- Job states
- Process suspension
- Process resumption
- Terminal control

6. FINAL VALIDATION

The project repository contains:

src/
build/
docs/
Makefile
README.md

Git repository:
Initialized successfully

Git tag:
v1.0

Working tree:
Clean

CONCLUSION
-----

The OSSP project was organized, built, tested, documented,
and versioned successfully.
j_rakshitha@RAKSHITHA:~/ossp_project$ git status
git tag
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    session24_task2_demo.txt

nothing added to commit but untracked files present (use "git add" to track)
v1.0
j_rakshitha@RAKSHITHA:~/ossp_project$ find . -maxdepth 2 -type f | sort
./git/COMMIT_EDITMSG
./git/HEAD
./git/config
./git/description
./git/index
./Makefile
./README.md
./build/builtin_shell
./build/child_exec
./build/child_process_demo
./build/double_quote_demo
```

j_rakshitha@RAKSHITHA: ~/o

```
./build/dquote_demo
./build/escape_demo
./build/escape_sequence_demo
./build/file_copy
./build/file_filter_pipe
./build/fork_exec
./build/history_shell
./build/keyboard_shell
./build/main
./build/memory_demo
./build/myshell
./build/parser
./build/path_resolver
./build/pipe_ls_grep
./build/pipe_producer_consumer
./build/pipeline_demo
./build/process_demo
./build/process_states
./build/session10_task1
./build/session9_task1_cd
./build/session9_task2_dispatch
./build/single_quote_demo
./build/squote_demo
./build/syntax_parser
./build/tokenizer
./build/variable_expand
./build/wait_waitpid
./build/waitpid_monitor
./build/zombie_demo
./copied.txt
./docs/session23_task2_architecture.txt
./docs/session23_task2_flow.txt
./sample.txt
./session24_task2_demo.txt
./src/builtin_shell.c
./src/child_exec.c
./src/child_process_demo.c
./src/double_quote_demo.c
./src/dquote_demo.c
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./src/keyboard_shell.c
./src/main.c
./src/memory_demo.c
./src/myshell.c
./src/parser.c
./src/path_resolver.c
./src/pipe_ls_grep.c
./src/pipe_producer_consumer.c
./src/pipeline_demo.c
./src/process_demo.c
```

```
./src/process_demo.c
./src/process_states.c
./src/session10_task1.c
./src/session10_task1.c.save
./src/session23_task1_pipeline.c
./src/session9_task1_cd.c
./src/session9_task2_dispatch.c
./src/single_quote_demo.c
./src/single_quotes.c+
./src/squote_demo.c
./src/syntax_parser.c
./src/tokenizer.c
./src/variable_expand.c
./src/wait_waitpid.c
./src/waitpid_monitor.c
./src/zombie_demo.c
j_rakshitha@RAKSHITHA:~/ossp_project$ git add session24_task2_demo.txt
j_rakshitha@RAKSHITHA:~/ossp_project$ git commit -m "Add final project demonstration"
[master 82573ee] Add final project demonstration
1 file changed, 88 insertions(+)
create mode 100644 session24_task2_demo.txt
j_rakshitha@RAKSHITHA:~/ossp_project$ git tag -f v1.0
Updated tag 'v1.0' (was a88ad7f)
j_rakshitha@RAKSHITHA:~/ossp_project$ git status
git tag
git log --oneline --decorate -2
On branch master
nothing to commit, working tree clean
v1.0
82573ee (HEAD -> master, tag: v1.0) Add final project demonstration
a88ad7f Initial OSSP project organization
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
j_rakshitha@RAKSHITHA: ~/o × + v
GNU nano 8.7.1 session24_task2_demo.txt
PRACTICAL SKILL SESSION 24 - TASK 2
=====

FINAL PROJECT DEMONSTRATION
=====

1. BUILD VERIFICATION

Command:
make

Result:
Project compiled successfully.

2. MAIN PROGRAM

Command:
./build/main

Result:
OSSP Project Started Successfully!

3. PIPELINE DEMONSTRATION

Program:
session23_task1_pipeline

Pipeline:
ls -l | grep .c | sort

Result:
Three-stage pipeline executed successfully.

4. SIGNAL DEMONSTRATION

The OSSP practical work included signal handling for:

SIGINT
SIGTERM
SIGUSR1
SIGTSTP
SIGCONT

Signals were tested with foreground and background processes.

5. PROCESS MANAGEMENT
```

The project work demonstrated:

- Process creation
- Process groups
- Foreground execution
- Background execution
- Job states
- Process suspension
- Process resumption
- Terminal control

6. FINAL VALIDATION

The project repository contains:

```
src/  
build/  
docs/  
Makefile  
README.md
```

Git repository:
Initialized successfully

Git tag:
v1.0

Working tree:
Clean

CONCLUSION

The OSSP project was organized, built, tested, documented,
and versioned successfully.

|